

DSD: Dual Self-Distillation for Robust Audio Tagging

Sen Wang, Feng Zhou, Zhicheng Zhang*, Zehao Wang, Jianqin Yin

School of Intelligent Engineering and Automation, Beijing University of Posts and Telecommunications, China

Email: {senwang, zhoufeng, zczhang, wzhao, jqyin}@bupt.edu.cn

Abstract—Knowledge Distillation (KD) is pivotal for enhancing the performance and compressibility of audio classification models. However, conventional KD typically relies on pre-training large-scale teacher models, significantly increasing training complexity and computational cost. In this paper, we propose a Dual Self-Distillation (DSD) framework tailored for Audio Tagging. This framework enables efficient knowledge transfer during the fine-tuning of pre-trained Vision Transformers (ViT) without external teachers, solely by harnessing the model’s internal potential. Specifically, to exploit internal representations, we introduce Hierarchical Intra-Layer Self-Distillation. This mechanism appends auxiliary classifiers to intermediate layers, utilizing deep semantic features and ground-truth labels to supervise shallow layers. Simultaneously, to improve robustness, we employ Evolutionary Mask-Consistency Distillation. This strategy combines the Mean Teacher mechanism with masked data augmentation, leveraging smoothed historical parameters to guide current training. Experiments on the DCASE 2025 Task 4 dataset demonstrate that DSD boosts baseline accuracy from 51.54% to 61.05%, verifying its effectiveness.

Index Terms—Audio tagging, knowledge distillation, self-distillation, consistency regularization, label augmentation

I. INTRODUCTION

Audio Tagging (AT) serves as a cornerstone in computational auditory scene analysis, enabling machines to interpret and categorize environmental sound events from complex acoustic mixtures. Historically, Convolutional Neural Networks (CNNs), represented by architectures like PANNs [1], dominated this field due to their translation equivariance and effectiveness in extracting local time-frequency patterns. However, the landscape of AT has recently undergone a significant transition with the introduction of Transformer into the audio domain. Models such as the Audio Spectrogram Transformer (AST) [2] and HTS-AT [3] have demonstrated that the self-attention mechanism outperforms CNNs by capturing long-range global dependencies, surpassing the limitations of local receptive fields. Consequently, leveraging Transformer-based backbones has become the standard for state-of-the-art audio classification.

In the AT task, balancing model performance with deployment feasibility under limited computational resources remains a critical challenge. Knowledge Distillation (KD) has emerged as an effective strategy to address this by transferring discriminative knowledge from large, high-performance models to lightweight student models [4]–[6]. The conventional teacher-student framework employs a powerful teacher

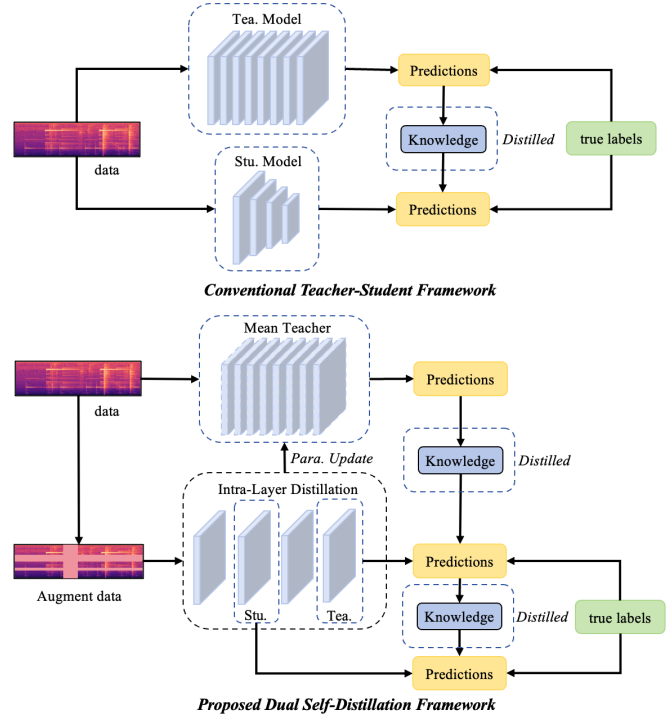


Fig. 1. Comparison of training paradigms. (Top) **Conventional Teacher-Student Framework**: Relies on an external, pre-trained teacher model to transfer knowledge, typically necessitating a complex two-stage training process. (Bottom) **Proposed Dual Self-Distillation (DSD)**: A single-stage framework that exploits the model’s intrinsic potential. It synergizes hierarchical (intra-layer) and evolutionary (mean teacher) self-supervision to achieve efficient training without any external teacher models. In this figure, Tea. denotes Teacher and Stu. denotes Student.

network to guide a smaller student via soft labels, enriching the semantic information learned by the student. Furthermore, AT task is often constrained by the scarcity of high-quality labeled data, which results in class imbalance and label noise. Training high-capacity models like Vision Transformers (ViT) on such datasets inevitably leads to overfitting. Consequently, data augmentation strategies are standard practices to enforce invariance and enhance robustness. Recent studies have further demonstrated that combining KD with consistent teaching under strong data augmentation [7] can significantly boost performance.

However, despite its widespread application, conventional KD faces multiple challenges in the audio domain. First,

* Corresponding author.

it requires pre-training a large teacher model before transferring knowledge, a two-stage process that significantly increases computational resource consumption. Moreover, mainstream KD paradigms typically involve heterogeneous architectures, such as employing large-scale Transformers to guide lightweight CNNs. This significant structural discrepancy creates inherent difficulties in aligning features across different spatial and semantic levels, thereby limiting the efficiency of knowledge transfer.

To overcome these limitations, we propose to mine the model’s intrinsic potential by distilling knowledge from itself rather than an external teacher. Accordingly, we introduce a single-stage training framework, Dual Self-Distillation (DSD), which integrates intra-layer self-distillation and mask-consistency self-distillation. We observe that current training paradigms often only provide supervision at the model’s final layer, failing to deeply harness the model’s intrinsic capabilities. Therefore, we abandon the traditional heterogeneous teacher-student paradigm and instead mine the model’s own potential. To resolve the inefficiency of feature learning, we design a Hierarchical Intra-Layer Self-Distillation mechanism. By inserting auxiliary classifiers after intermediate Transformer blocks, we force shallow layers to learn deep semantic features early, effectively bridging the semantic gap without external guidance. Furthermore, while data augmentation is essential for combating overfitting, indiscriminate masking can disrupt temporal continuity. To address this, we introduce Evolutionary Mask-Consistency Distillation by constructing an evolving teacher via the Exponential Moving Average (EMA) of model parameters. This teacher leverages stable predictions on unmasked data to guide the student’s learning on masked views. This synergistic supervision boosts the accuracy from 51.54% to 61.05%. The comparison between our proposed DSD framework and the conventional teacher-student framework is illustrated in Fig. 1.

In summary, the main contributions of this paper are as follows:

- Proposing a ViT-based Hierarchical Intra-Layer Self-Distillation mechanism that enforces deep-to-shallow semantic alignment. This strategy strengthens feature representation by utilizing internal supervision, eliminating the need for external teachers.
- Introducing an Evolutionary Mask-Consistency Distillation strategy with asymmetric masking. By using historical consistency to rectify predictions on masked data, we effectively improve model robustness against perturbations.
- Achieving a remarkable accuracy improvement of 9.51% on the DCASE 2025 Task 4 dataset, setting a new benchmark for single-model performance without external data.

II. RELATED WORK

A. Transformer-based Audio Tagging

Following the trajectory of Vision Transformers (ViT) [8], audio transformers have revolutionized Audio Tagging (AT).

The Audio Spectrogram Transformer (AST) [2] was among the first to apply patch-based processing directly to audio spectrograms, achieving superior performance over CNNs. To improve computational efficiency, HTS-AT [3] introduced a hierarchical token-semantic processing mechanism. More recently, self-supervised methods such as MAE-AST [9] and M2D [10] learn robust audio representations by reconstructing masked spectrogram patches, and can be effectively fine-tuned for downstream tasks.

B. Knowledge Distillation in Audio

Knowledge Distillation (KD) [11] is a widely adopted technique for transferring knowledge from a teacher network to a student network. In the audio domain, KD is extensively used for model compression, such as distilling heavy Transformers into efficient CNNs [6], [12]. Recent advancements, such as CED [4], have explored “consistent teaching” paradigms, where knowledge is distilled from large-scale ensembles using consistent data augmentation strategies. Additionally, Self-Distillation methods, exemplified by “Be Your Own Teacher” (BYOT) [13], propose distilling knowledge within a single network by utilizing deep layers to supervise shallow layers, thereby enhancing feature learning without external teacher models.

C. Consistency Regularization

Consistency regularization is a fundamental principle in semi-supervised learning, enforcing a model to output invariant predictions for perturbed versions of the same input. The Mean Teacher framework [14] implements this by maintaining an EMA of the model parameters to generate stable targets. This approach has been successfully applied in semi-supervised sound event detection [15] and label enhancement tasks, such as Pseudo Strong Labels (PSL) [5], to mitigate the impact of label noise and improve robustness against input perturbations.

III. METHOD

A. Framework Overview

We propose a DSD framework to enhance the performance of ViT-based audio tagging models. The core philosophy is to leverage the model’s intrinsic knowledge to regularize intermediate feature representations and improve generalization, without relying on external teacher networks. The framework integrates deep supervision, intra-layer knowledge distillation, and a mask-consistency strategy based on an EMA teacher.

Specifically, the training process incorporates two parallel distillation mechanisms. The first is Hierarchical Intra-Layer Distillation, where the deepest layer guides intermediate layers. We attach auxiliary classifiers to selected intermediate blocks, supervising them jointly with ground-truth labels and soft targets from the final layer to inject direct gradients into shallow blocks. The second is Evolutionary Mask-Consistency Distillation, utilizing historical knowledge to guide current learning. We employ an asymmetric strategy: the original audio serves as the weak view (x_w), while its masked version

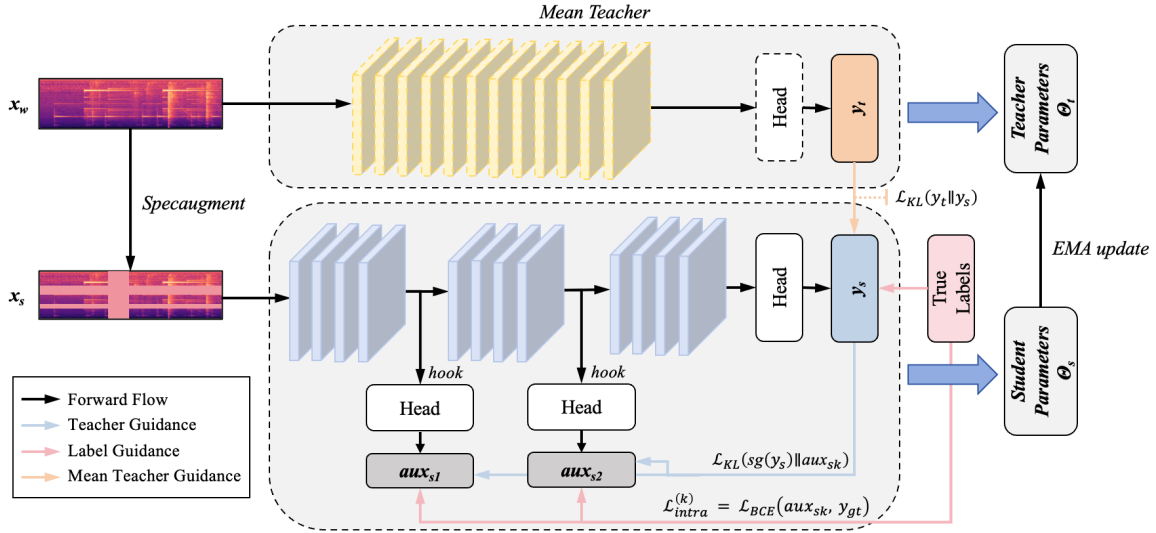


Fig. 2. Overview of the proposed Dual Self-Distillation (DSD) framework. The Student processes strongly augmented (masked) input x_s , while the EMA Teacher processes weakly augmented input x_w . The training is regularized by three synergistic objectives: (1) **Deep Supervision** (pink lines), injecting ground-truth signals into intermediate layers; (2) **Hierarchical Intra-Layer Self-Distillation** (blue lines), where the Student’s deep semantics guide shallow layers; and (3) **Evolutionary Mask-Consistency Distillation** (orange lines), where the Teacher’s stable prediction y_t rectifies the Student’s prediction under masking. **Notations:** $\mathcal{L}_{KL}(y_t || y_s)$: KL divergence loss from distillation (loss 2); $\mathcal{L}_{KL}(sg(y_s) || aux_{sk})$: KL divergence loss from distillation (loss 1); $\mathcal{L}_{intra}^{(k)} = \mathcal{L}_{BCE}(aux_{sk}, y_{gt})$: Binary Cross-Entropy loss from ground-truth labels.

serves as the strong view (x_s). The student model processes x_s to learn robust features under occlusion. Simultaneously, x_w is fed into a Mean Teacher, updated via EMA, to generate stable pseudo-labels that supervise the student’s prediction on the masked data.

B. Asymmetric Masking Strategy

To achieve effective mask-consistency distillation and enhance robustness against time-frequency noise, we construct two information-asymmetric views for each audio sample: a Weak View and a Strong View. Given the original Log-Mel spectrogram $x \in \mathbb{R}^{T \times F}$ (where T is the time dimension and F is the frequency dimension), the views are defined as follows:

Weak View (x_w): This view retains the complete structure of the original spectrogram without any time-frequency masking. It preserves full time-frequency semantic information and serves as the input to the Mean Teacher model. The goal is to generate high-quality, low-noise supervision targets based on complete information.

Strong View (x_s): Derived from the original spectrogram using the SpecAugment strategy. Specifically, multiple continuous regions are randomly selected along both the time and frequency axes and masked (set to zero). This simulates real-world scenarios such as frequency band loss or signal occlusion. x_s serves as the input to the Student model (the intra-layer distillation framework). This asymmetric strategy forces the Student to learn a prediction distribution consistent with the Teacher (which sees the full context) even when key semantic parts are missing, thereby compelling the model to leverage contextual associations to complete features and improve robustness.

C. Hierarchical Intra-Layer Distillation

In deep Transformer networks, intermediate layers often struggle to learn semantically discriminative features due to the long distance from the final supervision signal. To mitigate this, we introduce a hierarchical intra-layer distillation mechanism that propagates knowledge from the deepest layer back to intermediate layers.

We select a set of intermediate Transformer blocks K and attach auxiliary classification heads to them. To ensure feature compatibility, these heads mirror the final classifier’s architecture, consisting of a LayerNorm followed by a Linear layer. The training is driven by two complementary forces. First, we apply Deep Supervision using ground-truth labels y_{gt} , injecting direct gradient signals into shallow blocks to alleviate vanishing gradients. Second, we employ Deep-to-Shallow Soft Supervision, where the final prediction y_s serves as a soft target. Unlike conventional KD that relies on external teacher models, this strategy transfers the model’s own refined semantic knowledge from the deepest layer to earlier layers, bridging the internal semantic gap and enhancing feature alignment efficiently.

Formally, for the k -th intermediate layer, the loss function $\mathcal{L}_{intra}^{(k)}$ is defined as a combination of the binary cross-entropy loss against the ground truth and the Kullback-Leibler (KL) divergence against the final prediction:

$$\mathcal{L}_{intra}^{(k)} = \mathcal{L}_{BCE}(aux_{sk}, y_{gt}) + \gamma \mathcal{L}_{KL}(sg(y_s) || aux_{sk}) \quad (1)$$

Here, $sg(\cdot)$ denotes the stop-gradient operation, ensuring that the distillation process only updates the intermediate layers and auxiliary heads, preventing the final classifier from collapsing towards the intermediate representations.

D. Evolutionary Mask-Consistency Distillation

To further enhance the model’s robustness against noise and input perturbations, we incorporate an evolutionary consistency mechanism based on the Mean Teacher paradigm. The core idea is to enforce consistency between the student’s prediction on the masked view and the teacher’s prediction on the clean view.

The teacher parameters θ_t are updated at each training step as an EMA of the student parameters θ_s :

$$\theta_t \leftarrow \alpha\theta_{t-1} + (1 - \alpha)\theta_s \quad (2)$$

where α is a momentum coefficient (typically set to 0.999). This temporal averaging filters out the noise in the student’s stochastic updates, resulting in a more stable model. During training, we minimize the divergence between the teacher’s output on the weak view, $y_t = f(x_w; \theta_t)$, and the student’s output on the strong view, $y_s = f(x_s; \theta_s)$. The consistency loss is formulated as:

$$\mathcal{L}_{cons} = \mathcal{L}_{KL}(y_t \parallel y_s) \quad (3)$$

This objective encourages the student to produce predictions that are invariant to the masking applied to the input, effectively teaching the model to reconstruct missing contexts using the stable guidance from the teacher.

E. Total Optimization Objective

The final training objective is a weighted sum of the main classification loss, the intra-layer distillation loss, and the consistency loss. The total loss $\mathcal{L}_{total}$ is expressed as:

$$\mathcal{L}_{total} = \mathcal{L}_{BCE}(y_s, y_{gt}) + \lambda_{intra} \sum_{k \in K} \mathcal{L}_{intra}^{(k)} + \lambda_{cons} \mathcal{L}_{cons} \quad (4)$$

where λ_{intra} and λ_{cons} are hyperparameters balancing the contribution of the hierarchical and evolutionary distillation components. During the inference phase, the EMA teacher model and all auxiliary heads are discarded, and only the student backbone is retained. Thus, our method improves performance without incurring any additional computational cost or memory overhead during deployment.

IV. EXPERIMENTS

A. Experimental Setup

Dataset. All experiments are conducted by fine-tuning the official M2D pre-trained model on the DCASE 2025 Task 4 dataset, with no external data used. The dataset consists of synthetic audio mixtures simulating real-world SED scenarios: training/validation sets are generated on-the-fly with precise timestamps, while the test set is pre-synthesized under diverse acoustic conditions. Each 10-second clip contains 1–3 target events (SNR 5–20 dB) mixed with background noise and interference. We split the development set into training/validation subsets and use the main test subset (1,620 samples) for final evaluation, with accuracy as the main metric.

Baseline System. We adopt the official DCASE 2025 Task 4 baseline system [16], [17] as our backbone architecture.

TABLE I
COMPARISON WITH STATE-OF-THE-ART SYSTEMS ON DCASE 2025 TASK 4. "EXT. DATA" DENOTES USAGE OF EXTERNAL DATASETS. "ENS." DENOTES MODEL ENSEMBLING.

Model Configuration	Settings		Accuracy (%) $\uparrow$
	Ext. Data	Ens.	
<i>Baseline</i>			
Basic Baseline [16]	$\times$	$\times$	51.54
<i>Competitors</i>			
Qian_SJTU [19]	$\times$	$\checkmark$	47.72
Bando_AIST [20]	$\times$	$\checkmark$	49.51
Wu_SUSTech [21]	$\times$	$\times$	51.54
Choi_KAIST [22]	$\checkmark$	$\times$	55.80
Morocutti_CPJKU [23]	$\checkmark$	$\checkmark$	61.60
<i>Proposed</i>			
Ours (DSD)	$\times$	$\times$	61.05

Specifically, the audio tagging component utilizes the M2D-AS model [18], a specialized variant of the Masked Modeling Duo (M2D) framework. M2D is a self-supervised learning (SSL) foundation model pre-trained on AudioSet using a masked prediction objective. The M2D-AS variant is further fine-tuned on AudioSet (527 classes) by combining the SSL objective with a supervised classification loss, enabling it to extract robust, high-level audio representations. In our experiments, we utilize this pre-trained M2D-AS backbone to generate frame-level embeddings for downstream classification.

Implementation Details. Experiments are implemented using PyTorch Lightning. We employ an asymmetric view generation strategy: weak views undergo only spectrogram conversion, while strong views apply SpecAugment masking with a frequency mask parameter of $F = 24$ and a time mask parameter of $T = 192$. Following the Mean Teacher framework, we set the EMA momentum to $\alpha = 0.999$ to update the teacher model. The model is trained for 700 epochs using the AdamW optimizer (initial learning rate 10^{-5}) with a batch size of 16. Performance is assessed using the official DCASE 2025 evaluator.

B. Experimental Results

Main Results. We compare the proposed DSD framework against the official baseline and several top-performing systems from the DCASE 2025 Challenge. As detailed in Table I, these competitors adopt various strategies to enhance performance: Qian et al. [19] and Bando et al. [20] utilize model ensembling, while Choi et al. [22] incorporate external datasets. The top-performing competitor, Morocutti et al. [23], employs a combination of both strategies.

Our method achieves an accuracy of 61.05%, yielding a remarkable gain of 9.51% over the Basic Baseline. Crucially, our approach significantly outperforms systems that rely solely on ensembling (e.g., Qian, Bando) or external data (e.g., Choi, +5.25%). Furthermore, we achieve performance comparable to the state-of-the-art Morocutti system (61.60%). However, unlike Morocutti et al., who rely on a complex ensemble and

TABLE II
ABLATION RESULTS OF DISTILLATION STRATEGIES

Method	Configuration		Acc (%) $\uparrow$
	Intra-layer	Mean Teacher	
Basic Baseline	$\times$	$\times$	51.54
Intra-Layer Only	$\checkmark$	$\times$	58.06 ^{+6.52}
Mean-Teacher Only	$\times$	$\checkmark$	59.26 ^{+7.72}
Ours (Combined)	$\checkmark$	$\checkmark$	61.05 ^{+9.51}

TABLE III
ABLATION ON AUXILIARY LAYER POSITIONS

Strategy	Active Layers	Acc (%) $\uparrow$
No Aux. Layers	None	59.26
Shallow Focus	1, 2	59.38 ^{+0.12}
Distributed	3, 6, 9	59.01 ^{-0.25}
Dense Focus	1–11	59.44 ^{+0.18}
Deep Focus (Ours)	10, 11	61.05 ^{+1.79}

external supervision (AudioSet Strong), our result is achieved by a single model trained strictly on the official dataset without any external data. This verifies that fully exploiting the model’s intrinsic potential via DSD is a highly efficient and effective pathway to state-of-the-art performance.

C. Ablation Study

Impact of Auxiliary Layer Positions. We investigate where to apply auxiliary supervision in Table III. Results indicate that supervising deep layers (10, 11) is superior to shallow or dense supervision. Deep features are semantically closer to the final target, whereas forcing supervision on shallow layers disrupts the learning of low-level acoustic patterns.

Effectiveness of Distillation Strategies. Table II dissects the contribution of each component. Intra-layer distillation and Mean Teacher consistency bring relative gains of 6.52% and 7.72%, respectively. Their combination yields a synergistic effect, achieving the best performance (61.05%).

Data Augmentation Strategy. Table IV confirms the necessity of asymmetric augmentation. Providing the Teacher with clean data and the Student with masked data maximizes the Student’s capability to recover missing information, outperforming symmetric strategies.

Robustness Against Signal Occlusion. To evaluate resilience to data loss, we conducted a stress test using random Time Masking (0%–30%). As shown in Fig. 3, the proposed DSD framework exhibits superior robustness. First, the dual-distillation architecture intrinsically enhances stability: even the DSD w/o Masking variant retains 86.6% of its performance at 30% masking, significantly outperforming the Baseline’s 81.2% retention rate. Second, the Mask-Consistency strategy further elevates the performance envelope across all intensities. Most notably, at a 30% masking ratio, the full DSD model achieves 52.47%, surpassing the Baseline’s performance on unmasked data (51.54%). This confirms that our method ensures reliable operation even in adverse acoustic environments.

TABLE IV
ABLATION ON MASKING CONSISTENCY STRATEGIES

Strategy	Masking View		Acc (%) $\uparrow$
	Teacher	Student	
No Masking	$\times$	$\times$	57.78
Symmetric Masking	$\checkmark$	$\checkmark$	59.44 ^{+1.66}
Reverse Masking	$\checkmark$	$\times$	57.22 ^{-0.56}
Asymmetric (Ours)	$\times$	$\checkmark$	61.05 ^{+3.27}

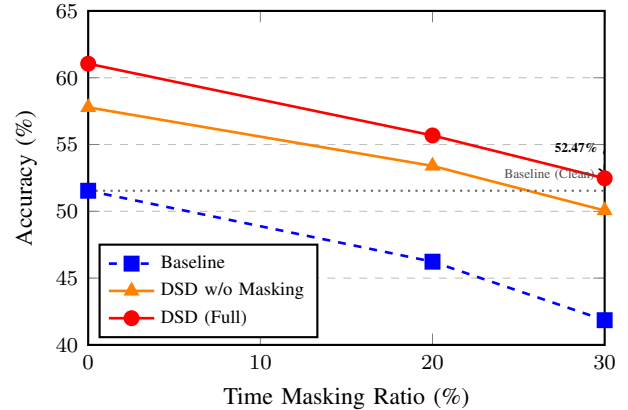


Fig. 3. Robustness evaluation under varying Time Masking Ratios (0-30%). The proposed DSD (Full, red) exhibits the highest resilience compared to the variant without masking (orange) and the Baseline (blue). Notably, at 30% masking, DSD retains 85.9% of its peak performance, whereas the Baseline retains only 81.2%, demonstrating superior robustness against signal loss.

D. Qualitative Analysis

Feature Representation Analysis. To validate the improvement in feature discriminability, we visualize the t-SNE embeddings of single-event samples from the deep feature space (Layer 10) in Fig. 4. As shown quantitatively, our method achieves a superior k-NN Accuracy of 0.94 compared to the Baseline’s 0.90. More importantly, the Silhouette Score sees a substantial improvement (0.33 vs 0.30), indicating significantly tighter intra-class clustering and clearer inter-class separation. Visual inspection confirms this metric gain: categories such as *Cough* (red circles) and *MechanicalFans* (orange diamonds), which are somewhat dispersed in the Baseline, form highly compact and well-separated clusters in our model. This demonstrates that the proposed dual self-distillation effectively regularizes the latent space, forcing the model to learn more robust and distinct semantic representations.

V. CONCLUSION

In this paper, we present the Dual Self-Distillation (DSD) framework, a single-stage training paradigm designed to maximize the model’s intrinsic potential for Audio Tagging. Instead of relying on computationally expensive external teachers, our approach synergizes Hierarchical Intra-Layer Distillation and Evolutionary Mask-Consistency Distillation to enforce deep semantic alignment and noise robustness from within. This

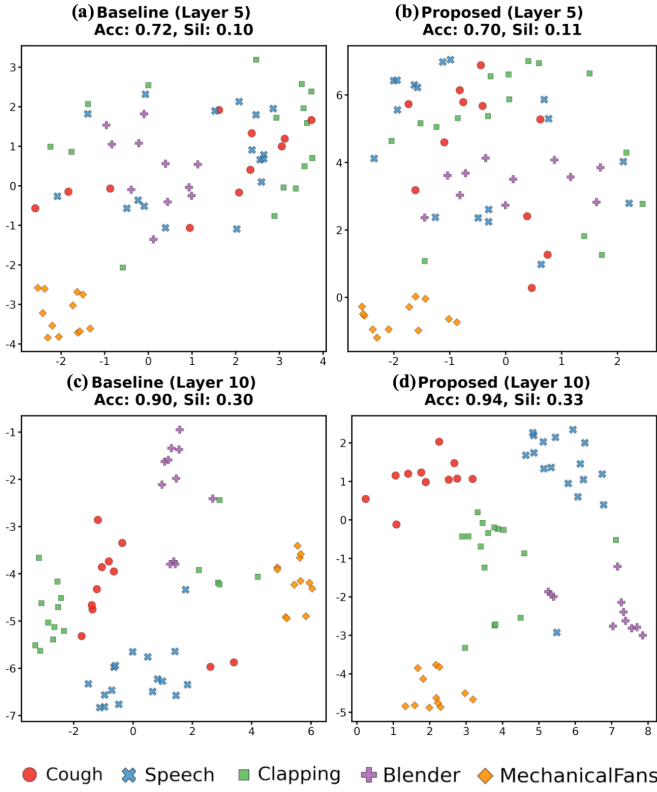


Fig. 4. t-SNE visualization of single-event samples. The four subplots correspond to: (a) Proposed (Layer 10), Acc: 0.94, Sil: 0.33; (b) Baseline (Layer 10), Acc: 0.90, Sil: 0.30; (c) Proposed (Layer 5), Acc: 0.70, Sil: 0.11; (d) Baseline (Layer 5), Acc: 0.72, Sil: 0.10. Compared to the baseline, the proposed method exhibits significantly tighter intra-class clustering and clearer inter-class separation in deep features (a vs. b). The marginal difference in intermediate layers (c vs. d) suggests that the semantic structure learned via deep supervision is progressively propagating to shallower layers.

strategy effectively transforms the model into its own teacher, ensuring efficient feature learning and consistent predictions even under strong data augmentation. Experimental evaluations confirm that DSD significantly outperforms conventional baselines, delivering superior classification accuracy and resilience in complex acoustic environments. Future work will explore applying this framework to other audio understanding tasks and investigating its effectiveness on larger-scale datasets.

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